Exercise 1 (SQL Fundamentals)

Question 1

```
--Question 1: Retreive all columns from the employees table
SELECT* FROM employees.table.employees;
```

Add parameter

Table

	employee_id	first_name	last_name	department	salary	hire_date
1	1	Alice	Green	IT	70000	2020-01-10
2	2	Brian	Lee	HR	45000	2019-03-22
3	3	Cathy	Zulu	Finance	65000	2018-07-18
4	4	David	Mokoena	Marketing	50000	2021-11-05
5	5	Eva	Naidoo	IT	72000	2017-09-30

Question 2

```
--Question 2: Find all unique departments
SELECT DISTINCT department
FROM employees.table.employees;
```

Add parameter

Table

	department
1	IT
2	HR
3	Finance
4	Marketing

Question 3

```sql
--Question 3: Retrieve first and last names ordered by salary descending
SELECT first_name,
       last_name
FROM employees.table.employees
ORDER BY salary DESC;
```

Add parameter

Table

	first_name	last_name
1	Eva	Naidoo
2	Alice	Green
3	Cathy	Zulu
4	David	Mokoena
5	Brian	Lee

Question 4

```sql
--Question 4: Retrieve the top 3 highest-paid employees
SELECT first_name,
       last_name,
       salary
FROM employees.table.employees
ORDER BY salary DESC
LIMIT 3;
```

Add parameter

Table

	first_name	last_name	salary
1	Eva	Naidoo	72000
2	Alice	Green	70000
3	Cathy	Zulu	65000

Question 5

```sql
--Question 5: Find employees in the IT department
SELECT employee_id,
       first_name,
       last_name,
       department
FROM employees.table.employees
WHERE department = 'IT';
```

Add parameter

Table

	employee_id	first_name	last_name	department
1	1	Alice	Green	IT
2	5	Eva	Naidoo	IT

Question 6

```sql
--Question 6: Find employees in Finance with salary > 60000
SELECT employee_id,
       first_name,
       last_name,
       department,
       salary
FROM employees.table.employees
WHERE salary > 60000 AND department = 'Finance';
```

Add parameter

Table

	employee_id	first_name	last_name	department	salary
1	3	Cathy	Zulu	Finance	65000

Question 7

```sql
--Question 7: Find employees in HR or Marketing
SELECT employee_id,
       first_name,
       last_name,
       department
FROM employees.table.employees
WHERE department = 'HR' OR department = 'Marketing';
```

Add parameter

Table

	employee_id	first_name	last_name	department
1	2	Brian	Lee	HR
2	4	David	Mokoena	Marketing

Question 8

```sql
--Question 8:Find employees not in IT
SELECT employee_id,
       first_name,
       last_name,
       department
FROM employees.table.employees
WHERE NOT department = 'IT';

--Question 9: Find employees in IT, HR or Finance using IN
```

Add parameter

Table

	employee_id	first_name	last_name	department
1	2	Brian	Lee	HR
2	3	Cathy	Zulu	Finance
3	4	David	Mokoena	Marketing

Question 9

```sql
--Question 9: Find employees in IT, HR or Finance using IN
SELECT employee_id,
       first_name,
       last_name,
       department
FROM employees.table.employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Add parameter

Table

	employee_id	first_name	last_name	department
1	1	Alice	Green	IT
2	2	Brian	Lee	HR
3	3	Cathy	Zulu	Finance
4	5	Eva	Naidoo	IT

Question 10

```sql
--Question 10: Find employees in IT with salary > 65000 and city Johannesburg
SELECT employee_id,
       first_name,
       last_name,
       department,
       salary,
       city
FROM employees.table.employees
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

Add parameter

Table

	employee_id	first_name	last_name	department	salary	city
1	1	Alice	Green	IT	70000	Johannesbu...
2	5	Eva	Naidoo	IT	72000	Johannesbu...